Forecasting Bitcoin Volatility Using Regime-Aware Machine Learning Models

DSE4211 Final Report

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1 Introduction

Cryptocurrency markets are widely recognised for their exceptional volatility and rapidly evolving structure. Unlike traditional financial markets, cryptocurrencies trade continuously, respond almost instantaneously to news and sentiment, and are heavily influenced by speculative activity (Fang et al., 2025). A growing body of literature documents several stylised facts, including pronounced volatility clustering, frequent jumps, structural breaks, and strong time variation in return dynamics (Charles et al., 2019; Bouri et al., 2019). These characteristics suggest that cryptocurrency volatility does not follow a stable, time-invariant process, but instead evolves across distinct market states.

Despite this, much of the volatility forecasting literature continues to rely on single-regime models that assume constant parameters over time. This raises an important empirical question: if cryptocurrency markets are inherently regime-dependent, does explicitly modelling regime shifts lead to superior forecasting performance?

Motivated by this gap, our study investigates whether a two-stage regime aware framework improves daily Bitcoin volatility forecasts relative to single-regime benchmarks. By comparing regime-aware and standard models using out-of-sample evaluation metrics, we aim to assess whether incorporating regime dynamics provides meaningful gains in predictive accuracy in the one-day-ahead forecast of Bitcoin's volatility. We evaluate three regime-unaware benchmark models, GARCH, EGARCH, and SVR-L, against three regime-aware models, HMM-SVR-L, RF-SVR-L, and LSTM-SVR-L, where each regime-aware model incorporates a forward-looking regime probability derived from a stage-one classifier into a common SVR-L forecasting stage.

1.1 Literature Review

Hashish et al. (2019) propose a hybrid model for Bitcoin price prediction combining Hidden Markov Models and Long Short-Term Memory networks. A two-state model was found to achieve the highest data likelihood, with the decoded state sequence used as an additional feature passed into a LSTM for price prediction. Their proposed hybrid model outperformed both ARIMA and LSTM across MSE, RMSE, and MAE on the out-of-sample test set, demonstrating

that regime information extracted by the HMM meaningfully improves predictive performance when combined with a sequential forecasting model. This approach is supported by Ardia et al. (2019), which demonstrated that regime changes is present in Bitcoin's volatility dynamics. By using Markov-switching GARCH (MSGARCH) models, the paper found that accounting for these shifts significantly improves Value-at-Risk (VaR) forecasts compared to single-regime models. Hence, these papers provide a theoretical justification for the use of regime-aware architectures in cryptocurrency markets.

2 Data Description / Data Processing

2.1 Data sources

The primary dataset consists of Bitcoin price data from the Coinbase API, including hourly and daily open, high, low, close prices, and trading volume for BTC–USD. These data are used to construct return and volatility measures. To capture blockchain fundamentals, on-chain metrics from the Blockchain.com Charts API, such as transaction counts and mempool activity, are included to reflect network demand and congestion. Macroeconomic and financial variables are obtained from Yahoo Finance via the `yfinance` library, including equity indices (S&P 500, Nasdaq, Dow Jones, Russell 2000), commodities (gold, silver, Brent crude oil), exchange rates (USD/EUR, USD/CNY), and the VIX index, capturing broader market conditions and risk sentiment. Investor attention is proxied using Google Trends data via the `pytrends` library, measured by search intensity for the keyword "Bitcoin.". Additional sentiment indicators, such as the Fear and Greed Index, were initially considered but eventually excluded due to substantial missing data.

2.2 Data Processing and Feature Engineering

All datasets were standardised to a daily frequency and merged using the date as the primary key. Missing values, primarily arising in financial indicators due to non-trading days, were addressed using forward filling to preserve temporal continuity and ensure consistency in the time series.

$$RV_t = \sum_{h=1}^{24} r_{t,h}^2 \tag{1}$$

Following Dudek et al. (2024), daily realized variance is defined as the sum of squared intraday (hourly) log returns, as shown above. Three lags and a 30-day rolling mean of realized variance were constructed to capture both short-term dependence and medium-term volatility persistence, as evidenced by the ACF and PACF structures (see Figure 2 in Appendix). Additionally, we model the logarithm of realized variance rather than the raw variance to stabilise the variance and reduce skewness (See Figure 1 in Appendix).

Several variables, including trading volume and blockchain activity measures, were also transformed using logarithms for similar reasons. For variables containing zero values, a small constant was added prior to transformation to ensure numerical stability. Additionally, first differences of log-transformed blockchain variables were computed to represent growth rates, capturing changes in network activity rather than levels, and reducing non-stationarity. See Table 6 for summary statistics.

Similarly, macroeconomic and cross-asset variables were expressed in terms of returns and volume changes, ensuring consistency in scale and interpretation across features. The final sample comprises daily observations from 13 July 2017 to 31 January 2026. See Table 4 and Table 5 in Appendix for the full feature set and transformations applied.

3 Methodology and Model Evaluation

The sample comprises 3,125 observations, split into 60/25/15 for the baseline (single-stage) model. For the two-stage framework, the evaluation set is further divided into two equal parts, while the training and test sets remain unchanged. The first subset is used to tune the regime model, and the second to tune the forecasting model using regime probabilities generated from the fixed first stage. This design prevents overly optimistic performance from joint tuning and reduces the search space. Model selection is based on the QLIKE loss, which balances over- and under-prediction and is less sensitive to outliers. The full set of hyperparameters is reported in Tables 8, 9, and 10 in the Appendix.

All models are estimated using a fixed rolling window of 365 observations, consistent with

the daily forecasting frequency and the focus on recent market dynamics. This choice follows Vinícius Veloso (2025), who demonstrate that a one-year window captures both short-term shifts and longer-term movements, making it a suitable benchmark for Bitcoin volatility forecasting.

3.1 Linear Support Vector Regression (SVR-L)

The baseline forecasting model employs a Linear Support Vector Regression (LinearSVR) to predict next-day Bitcoin realized variance. The selection of the SVR-L is motivated by its consistent performance in cryptocurrency literature (Dudek et al., 2024), where it frequently outperforms traditional econometric and complex deep learning models like LSTM in terms of MSE and MAE. As a regime-unaware model, the SVR-L operates under a global optimization framework, assuming that the relationship between input features and the target variable remains constant across the entire estimation window.

The model is trained to minimize the epsilon-insensitive loss function, seeking a hyperplane that fits the majority of the observations within a specified margin. This baseline serves as the benchmark to evaluate the incremental predictive value of the derived regime probabilities in the second stage of the integrated model.

3.2 Regime-Aware Models

Bendtsen (2016) characterises a regime as a period in which the probabilistic relationships and distributions of observed variables remain stable within a given latent state. Regime-aware models are designed to capture transitions between these states, allowing for structural changes in the underlying statistical properties across regimes. Consequently, the development of a regime-aware model architecture requires a formal definition of the latent states, which can either be specified a priori based on domain knowledge or endogenously determined from the data using unsupervised learning techniques. We adopt both approaches for our investigation, employing a Hidden Markov Model for unsupervised state identification and following Agakishiev et al. (2025) for the supervised specification. For tractability, we restrict the analysis to two regimes (low and high) so as to mitigate class imbalance issues that may arise with a larger number of regimes (see Table 7 in Appendix). Low volatility regime is characterised

by relatively stable and predictable price movements, whereas high-volatility regime features larger fluctuations and higher risk for investors. The choice of a two-regime specification is further supported by the findings of Joshi (2025), who show that a two-regime model provides a statistically stable separation of the data into distinct low- and high-volatility states. In contrast, models with a larger number of regimes tend to produce states that are either statistically insignificant or redundant due to insufficient observations. Taken together, both the existing literature and our empirical findings motivate the focus on a binary classification framework.

With the states identified, we construct a two-stage model architecture. In the first stage, a regime classification model estimates the probability that the market will be in regime k at time $t+1$, conditional on information available at time t. In the second stage, these predicted regime probabilities, along with other covariates, are incorporated into a variance forecasting model to predict next-period realized variance, given by the following specification:

$$\mathbf{j}_{t+1} = (P(S_{t+1} = k \mid X_t))_{k=1}^{K}, \quad \mathbf{j}_{t+1} \in \Delta^{K-1} \tag{2}$$

where S_{t+1} denotes the latent regime at time $t+1$, X_t is the input feature vector at time t, and $\mathbf{j}_{t+1}$ is the vector of predicted regime probabilities over the K possible regimes. The predicted regime probabilities are then incorporated into the variance forecasting equation:

$$\log(RV_{t+1}) = f(\beta_0 + \boldsymbol{\beta}'\mathbf{Z}_t + \boldsymbol{\gamma}'\mathbf{j}_{t+1} + \varepsilon_{t+1}) \tag{3}$$

where β_0 is an intercept, $\boldsymbol{\beta}$ is a vector of coefficients on the covariate vector $\mathbf{Z}_t$, $\boldsymbol{\gamma}$ is a vector of regime loadings, and ε_{t+1} is an error term. In our framework, f is specified as a linear SVR model.

3.2.1 Hidden Markov Model and SVR-L

Stage 1: Regime Probability Labelling with Hidden Markov Model. The first stage employs a two-state Gaussian Hidden Markov Model (HMM) to identify latent volatility regimes in Bitcoin realized variance. Following Hashish et al. (2019), the HMM input features include log returns and log rolling realized variance over 1-day and 30-day windows. These variables are standardised using a zero-mean, unit-variance scaler before being passed into the model.

The HMM assumes that the observed feature sequence is generated by one of two hidden states: a low-volatility regime and a high-volatility regime. Each state is characterised by a multivariate Gaussian distribution with a diagonal covariance matrix. Model parameters are estimated using the Baum–Welch algorithm, while the most likely sequence of hidden states is decoded using the Viterbi algorithm. To ensure consistency across rolling estimation windows, the estimated states are relabelled according to mean realized variance, such that state 1 always corresponds to the high-volatility regime and state 0 to the low-volatility regime.

The HMM is estimated using a rolling 365-day window and refitted at each time step t using only past observations. The fitted transition matrix is then applied to the filtered state distribution at time t to obtain the predicted probability of being in the high-volatility regime at $t + 1$, denoted by `hmm_prob_high_t1`. This probability is subsequently used as an input to the second-stage SVR model without introducing look-ahead bias.

Stage 2: Variance Forecasting with Linear SVR. The HMM–SVR-L model extends the baseline SVR-L specification by including `hmm_prob_high_t1` as an additional predictor. This allows the forecasting model to condition on the next day's expected volatility regime, rather than relying solely on lagged and contemporaneous market features. The same data split and hyperparameter tuning procedure used in the baseline SVR-L model is applied.

3.2.2 Supervised Regime Detection

Volatility Regime Labelling. Following Agakishiev et al. (2025), regimes are identified using a rolling percentile-based approach. At each time step, the percentile rank of realized volatility over the past 30 days is computed. Observations below the 30th percentile are classified as low-volatility regimes, while observations above the 60th percentile are classified as high-volatility regimes. To prevent excessive regime switching, a buffer zone is introduced such that the regime remains unchanged unless the threshold for a new regime is clearly crossed. These labels are then used as target variables for the supervised regime classification models.

Stage 1: Supervised Regime Detection. We employ two machine learning models to predict the subsequent-period volatility regime: Long Short-Term Memory (LSTM) networks and Random Forests (RF). LSTM networks are used due to their ability to model sequential dependence in time-series data and their role as a benchmark architecture within recurrent neural networks

(Giantsidi and Tarantola, 2025). Random Forests are also considered due to their strong predictive performance and robustness to overfitting through ensemble learning (Breiman, 2001).

Following Agakishiev et al. (2025), we also implement a weighted LSTM objective, where observation-specific weights are introduced into the loss function to place greater emphasis on regime-switching observations. We set $\lambda = 0.5$ and provide the full weighting scheme in Appendix 6. Both the LSTM and RF models are trained using features capturing market, macro-financial, blockchain, and investor attention dynamics.

Stage 2: Variance Forecasting with Linear SVR. The second-stage forecasting model uses a Linear SVR specification. Unlike the first-stage regime classifiers, the SVR adopts a more parsimonious feature set, relying primarily on lagged realized variance, recent volatility history, and the predicted regime probabilities from Stage 1. This design mitigates overfitting by allowing the first-stage models to compress high-dimensional information into informative regime signals, which are then used to condition the variance forecast.

4 Model Performance Evaluation and Analysis

GARCH(1,1) was used as the baseline for comparison against the single and double regime models while EGARCH was excluded (See Limitations and future works).

Table 1: Out-of-sample forecasting performance across models

Model	RMSE	MSE	MAE	QLIKE
GARCH	0.000652	4.24×10^{-7}	0.000403	0.433337
SVR–L	0.000618	3.82×10^{-7}	0.000383	0.411096
LSTM–SVR	0.000666	4.44×10^{-7}	**0.000368**	0.525050
RF–SVR	**0.000609**	$\mathbf{3.71 \times 10^{-7}}$	0.000374	**0.385672**
HMM–SVR	0.000646	4.18×10^{-7}	0.000388	0.451964

According to Table 1, regime-aware models generally outperform regime-unaware models across all loss-based metrics, although the magnitude and statistical significance of improvements vary by specification. For the regime-unaware models, SVR-L emerges as the stronger performer, suggesting that even a single-regime machine learning model provides a meaningful improvement over the traditional econometric specification.

Incorporating regime information leads to selective but economically meaningful gains. The RF-SVR model achieves the lowest RMSE, MSE and QLIKE while the HMM-SVR model

delivers modest improvements in point forecast metrics. These results suggest that regime-aware approaches enhance predictive accuracy, with machine learning-based classifiers providing greater benefits than the traditional HMM framework. The LSTM-SVR model presents a more nuanced performance profile. While it does not outperform RF-SVR in RMSE or MSE, it achieves the lowest MAE , indicating superior performance in minimising average absolute forecast errors. However, its relatively high QLIKE suggests that its variance forecasts may be less well-calibrated, particularly in periods of elevated volatility.

Table 2: Diebold–Mariano (DM) test statistics with GARCH as the benchmark

Benchmark	**SVR**	**LSTM–SVR**	**RF–SVR**	**HMM–SVR**
GARCH	1.603	-1.008	**2.169**	0.297

Note: The table reports Diebold–Mariano test statistics comparing predictive accuracy relative to the benchmark model. Bolded values indicate rejection of the null hypothesis of equal predictive accuracy at 5% significance level.

Comparing across models in Table 2, it can be observed that not all regime-aware models improve upon GARCH at 5% level. While RF-SVR delivers clear and significant gains, LSTM-SVR underperforms the benchmark. HMM-SVR, while economically superior across most metrics, is statistically indistinguishable compared to GARCH.

Table 3: Diebold–Mariano (DM) test statistics with RF–SVR as the benchmark

Benchmark	**LSTM–SVR**	**HMM–SVR**
RF–SVR	-2.450	-1.593

Note: The table reports Diebold–Mariano test statistics comparing predictive accuracy relative to the benchmark model. Bolded values indicate rejection of the null hypothesis of equal predictive accuracy at the 5% significance level.

From Table 3, the negative and statistically significant Diebold–Mariano statistic indicates that RF–SVR delivers superior forecast accuracy compared to LSTM–SVR. This may be partly attributed to RF–SVR's ability to better capture volatility spikes, whereas LSTM–SVR appears to underpredict extreme movements (See Figure 3 in Appendix). This likely reflects RF's strength in modelling nonlinear interactions and abrupt regime shifts, while LSTM may smooth over sharp transitions and struggle to adapt to sudden shocks in the data. In contrast, while the DM statistic comparing RF–SVR and HMM–SVR is also negative, it is not statistically sig-

nificant at the 5% level, indicating that the null hypothesis of equal predictive accuracy cannot be rejected. As such, although RF–SVR exhibits advantages associated with more flexible data-driven regime identification, its performance advantage over HMM–SVR is not robust in a statistical sense.

4.1 Regime classification

Table 11 in Appendix shows the classification report for the LSTM and RF model classification. Overall, both models achieve comparable performance, with the LSTM attaining a slightly higher accuracy of 0.61 compared to 0.59 for the Random Forest. The LSTM specification also exhibits more balanced performance across regimes, with relatively similar precision, recall, and F1-scores for both low- and high-volatility states, whereas the Random Forest shows a tendency to under-identify high-volatility periods.

Feature importance analysis (see Figure 4 in the Appendix) provides further insight into the differences between the two models. The LSTM model relies heavily on on-chain metrics for regime identification, consistent with Agakishiev et al. (2025), with the growth rate of unique addresses emerging as the most influential predictor. This suggests that shifts in network participation may act as forward-looking signals for regime transitions. In contrast, the Random Forest primarily captures autoregressive and contemporaneous dynamics, behaving more similarly to classical econometric models that rely on past values and current market conditions.

4.1.1 Error Propagation in the Two-Stage Framework

One of the key concerns in a multi-stage architecture such as this is the potential for error propagation across stages, whereby inaccuracies in the first-stage model are transmitted and potentially amplified in subsequent stages. To evaluate this, we construct two additional benchmarks. First, an oracle model, where the true regime label is used as input, representing a best-case scenario with no first-stage error. Second, a neutral baseline, where regime probabilities are fixed at 0.5 for all observations, effectively removing any regime information. By comparing forecasting performance of the regime-aware model with these two additional models, we are able to isolate both the value of regime information and the cost of estimation error.

Table 12 in the Appendix describes the results and reveal three key insights. First, the oracle

model achieves the best forecasting performance across all metrics, indicating that regime information contains meaningful predictive signal for next-day realized volatility. Second, most regime-aware models, particularly the LSTM–SVR and HMM–SVR specifications, underperform the neutral baseline. This suggests that the predicted regime probabilities from these models introduce substantial noise, which propagates into the second-stage model and ultimately outweighs the potential benefits of incorporating regime information into volatility forecast. Third, the RF–SVR model performs comparably to the neutral baseline, indicating that not all first-stage models are equally affected by this issue. This highlights that the effectiveness of the two-stage framework is highly sensitive to the quality and stability of the regime estimates.

The distribution of predicted regime probabilities (see Figure 6 in Appendix) provides further insight into the observed performance differences between the LSTM and RF specifications. The LSTM model produces highly concentrated probabilities at the extremes, with most predictions close to 0 or 1. This suggests that the model is overconfident and effectively reduces the probabilistic output to a near-deterministic classification. Moreover, the scatter plot indicates that these predictions are only weakly related to realized volatility, with a correlation of approximately 0.197. As a result, the regime signal is both noisy and poorly aligned with the forecasting target. Consequently, the second-stage model is exposed to strong but unreliable signals, leading to significant error propagation. Conversely, the Random Forest model produces a more dispersed and better-calibrated distribution of probabilities, and exhibits a more stable relationship with realized volatility, with a higher correlation of approximately 0.318. This is likely the reason why the RF–SVR specification performs comparably to the neutral baseline, whereas the LSTM–SVR model underperforms.

The HMM-based specification suffers from the same problem. It was refitted on a rolling window using only three input features, making its state estimates heavily reliant on recent volatility levels. When volatility is transitioning, the Gaussian emission model with a fixed diagonal covariance may misclassify regimes, as it lacks access to the broader macro-financial and blockchain feature set available to the supervised models. As a result, the regime probabilities produced by the HMM are poorly aligned with the forecasting target, introducing noise that propagates into the second stage and offsets any potential gains from regime conditioning.

The results provide strong evidence of error propagation: while regime information is valuable in principle, its practical usefulness depends critically on the reliability of the first-stage predictions. A classification accuracy of around 60% may be insufficient for the regime signal to be beneficial in the downstream forecasting task. However, classification accuracy alone is only one component; the stability, calibration of predicted regime probabilities, are equally important.

5 Conclusion

Overall, while all model specifications tend to underestimate volatility spikes, we observe clear heterogeneity in performance among regime-aware models. In particular, the Random Forest specification outperforms other models with statistical significance, suggesting that incorporating regime information can provide useful signal for volatility forecasting. However, error propagation in a multi-stage architecture remains a key concern. Despite its complexity, the Random Forest model performs only comparably to a neutral regime specification in the downstream forecasting task. Furthermore, results from the LSTM-based regime prediction indicate that better first-stage classification does not automatically translate into improved overall performance, as inaccuracies and instability in the estimated regime probabilities can offset the potential benefits of incorporating regime information. Therefore, our findings highlight that the effectiveness of regime-based approaches depends not only on predictive accuracy, but also on the reliability and downstream relevance of the estimated regime signals. Additionally, our results show that the gains from using regime probabilities as a compressed representation of the broader information set appear limited. Therefore future work could explore alternative approaches, such as principal component analysis (PCA), to more effectively summarise high-dimensional information.

5.1 Limitations and future works

One of the limitations we faced was the failure of EGARCH to converge during estimation. EGARCH was initially selected as one of the single-regime models and as the second step for the double-regime models. However, due to multiple 'explosions' in the predictions even during the validation stage (see Figure 7 in Appendix), EGARCH was unable to converge

and produced unsatisfactory results in terms of evaluation metrics. Hence, we decided to use GARCH as the baseline model.

Second, our preliminary empirical experiments indicate that a naive regime-switching implementation, where two separate GARCH models are estimated on subsamples defined by a binary regime filter, does not yield meaningful improvements in forecast performance. This approach likely disrupted the inherent intertemporal dependence structure of the volatility process, and GARCH models rely on such continuous lagged dynamics which are no longer preserved under such segmentation. Therefore, future work could explore more coherent regime-switching frameworks that preserve the intertemporal dependence structure such as other non-linear approaches that incorporate regime probabilities in a continuous manner.

Third, while Markov-Switching GARCH (MSGARCH) was considered to address regime instability, preliminary testing revealed that the model's accuracy was overly sensitive to the precision of the underlying Markov transition probabilities. Inaccurate probability assignments often resulted in the premature application of variance multipliers, leading to artificial volatility spikes. Future research should therefore focus on Bayesian MSGARCH frameworks using Dirichlet Process priors to 'smooth' these transitions, or alternatively, the implementation of Neural Gated GARCH models. By utilizing a continuous gating mechanism (such as a GRU or TCN) instead of discrete Markov states, the model could achieve a more fluid and numerically stable adaptation to shifting market regimes.

Lastly, future work could explore the sensitivity of results to alternative rolling window lengths, such as 180 or 730 days, to assess whether the findings are robust to different estimation horizons given Bitcoin's structural breaks over time. Extending the framework to other cryptocurrencies would also help evaluate the generalisability of the regime-aware approach beyond Bitcoin.

6 Rebuttal

Several reviewers noted that the literature review was insufficiently developed, with limited engagement beyond a single study. In response, we have expanded this section by incorporating additional relevant work, most notably Ardia et al. (2019), which demonstrates the presence of regime shifts in Bitcoin volatility using a Markov-switching GARCH framework.

Concerns were also raised regarding the treatment of EGARCH, which was introduced as a benchmark but omitted from the results without early explanation. We have now addressed this by explicitly flagging at the beginning of the Model Performance Evaluation and Analysis section that EGARCH was excluded due to convergence issues, rather than deferring this explanation to the Limitations and Future Work section. This ensures greater transparency in the model selection process.

The choice of a two-regime specification was another area highlighted by reviewers as lacking empirical justification. To address this, we have included additional analysis comparing class distributions under two- and three-regime specifications, with results reported in Table 7 of the Appendix.

Relatedly, the choice of a 365-day rolling window was flagged as potentially arbitrary. We have strengthened the justification for this choice by referencing Vinícius Veloso (2025), who shows that a one-year window strikes a balance between capturing short-term fluctuations and longer-term structural dynamics in Bitcoin markets. This provides a clearer rationale for our rolling estimation framework.

One reviewer raised an important concern regarding the relatively modest regime classification accuracy (approximately 58–60%). In response, we have added a dedicated discussion under the Model Performance Evaluation and Analysis section examining whether predictive gains arise from genuine regime detection or merely from proxying recent volatility levels.

Furthermore, we provide a dedicated explanation for the underperformance of the HMM specification, attributing it to its limited feature set and the constraints of the Gaussian emission model, which lacks access to the broader macro-financial and blockchain information available to the supervised models.

Reviewers also noted that the supervised regime classification thresholds appeared asymmetric and insufficiently justified. We clarify in the final report that these thresholds are adopted directly from Agakishiev et al. (2025), rather than being chosen heuristically, ensuring consistency with the underlying methodological framework.

A major methodological concern related to inconsistencies in the feature sets used across second-stage SVR models, which undermined the validity of controlled comparisons. We acknowledge this limitation and partially address it by standardising the feature sets across the LSTM-SVR and RF-SVR specifications. While the HMM-SVR framework differs in how regime probabilities are generated, all models are constructed from the same underlying information set. In the two-stage frameworks, a richer set of covariates is used in the first stage to extract regime probabilities, which are then incorporated into a more parsimonious second-stage SVR model. This reflects a deliberate modelling choice, allowing us to assess whether regime probabilities can also serve as a compressed representation of higher-dimensional information, rather than relying solely on a large set of predictors in the final forecasting stage. As such, the results should be interpreted not only as evidence on regime incorporation, but also in terms of the trade-off between model complexity and information compression. Another methodological concern raised by the reviewers relates to differences in the re-estimation of model parameters across the machine learning models and the HMM specification. We acknowledge this concern, but we feel that re-training machine learning models at every time step would be computationally expensive and may introduce additional instability in model estimation. Furthermore, if anything, this asymmetry is likely to disadvantage the machine learning specifications but we do not observe systematically poorer performance relative to the HMM. In fact, the Random Forest–SVR model delivers the strongest performance among the regime-aware specifications, suggesting that the effectiveness of the two-stage framework depends more critically on the quality of the estimated regime probabilities than on the re-estimation scheme itself. Nevertheless, consistent re-estimation framework of all models would provide a more symmetric comparison.

Another technical issue was identified in the implementation of the Diebold–Mariano test. Specifically, the use of multiple lags in the Newey–West estimator was inappropriate for one-

step-ahead forecasts. Following the reviewer's suggestion, we have revised the implementation to use zero truncation lags after confirming the absence of autocorrelation in forecast errors. This correction improves the reliability of our statistical inference.

In addition, a plotting error involving double exponentiation in the SVR outputs was identified. While this did not affect the reported evaluation metrics, it distorted the visualisation of variance forecasts. This issue has been corrected in the final report. Reviewers also noted that the actual versus predicted variance plot was visually difficult to interpret due to the overlay of multiple model outputs in a single chart. In response, we have restructured the figure into two separate panels, with the first displaying regime-unaware models (GARCH and SVR-L) and the second displaying regime-aware models (LSTM-SVR, RF-SVR, and HMM-SVR), each plotted against actual realized variance. This separation allows for clearer visual comparison both within and across model families (seen in Figure 3 in the Appendix).

Beyond visualisation concerns, reviewers also noted that Google Trends captures attention magnitude but not sentiment direction, and that a HAR model would serve as a more natural baseline given that it directly targets realized variance. We acknowledge both points. Regarding sentiment, future work could incorporate news-based or social media indicators for a more directionally informative signal. Regarding the HAR baseline, we note that the lagged realized variance terms included in the SVR-L specification subsume the core autoregressive structure of the HAR model, partially mitigating this concern. Due to page constraints, these points are addressed here rather than in the main report.

Finally, several reviewers commented on the organisation and usability of the project repository. In response, we have added a comprehensive README file and a requirements.txt to document dependencies and execution steps. We have also removed redundant notebooks, stale files, and unused datasets, streamlining the repository to include only essential components. Furthermore, each notebook now begins with an overview section to support reader accessibility.

Overall, these revisions have improved the clarity, robustness, and reproducibility of the study. We are grateful to the reviewers for their insightful feedback, which has significantly strengthened the final report.

Appendix

Table 4: Data Sources and Variables

Source	**Variable**	**Description**
Market Data (Coinbase API)		
close (daily)	Daily closing price	Used to compute returns
close (hourly)	Hourly closing price	Used to compute realized variance
volume	Daily trading volume	Market activity proxy
Blockchain Data (Blockchain.com)		
n_transactions	Number of transactions	Network usage
transaction_fee_usd	Transaction fee (USD)	Transaction cost
n_unique_addresses	Active addresses	Network activity
transaction_volume	Transaction volume (USD)	Economic activity
median_confirmation_time	Confirmation time	Network congestion
mempool_count	Pending transactions	Network backlog
hash_rate	Hash rate	Network security
difficulty	Mining difficulty	Computational effort
Financial Data (Yahoo Finance)		
gold, silver, brent	Commodity prices	Market conditions
dji, spx, rut, nasdaq	Equity indices	Market performance
usdcny, usdeur	Exchange rates	FX market
vix	Volatility index	Market uncertainty
Google Trends		
google_trend	Search interest	Investor attention

Table 5: Variable Transformations

Variable Type	**Transformation**
Prices	Log returns
Hourly prices	Realized variance
Volumes	Log transformation / percentage change
Blockchain activity variables	Log-differences
Network variables (hash rate, difficulty)	First differences
VIX	First difference
Google Trends	Percentage change

Table 6: Summary Statistics of Model Variables

Variable	Mean	Std	Min	Max
log_realized_variance	-7.456	1.343	-11.959	-2.392
log_return	0.001	0.036	-0.491	0.241

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Variable	Mean	Std	Min	Max
gtrend_pct_change	0.015	0.191	-0.680	2.760
blockchain_diff_log_n_transactions	0.000	0.145	-0.843	0.769
blockchain_diff_log_transaction_fee_usd	0.000	0.274	-2.309	2.485
blockchain_diff_log_n_unique_addresses	0.000	0.113	-0.479	0.426
blockchain_diff_log_transaction_volume_usd	0.001	0.517	-1.773	2.384
log_volume	9.314	0.725	6.529	11.778
gold_close_ret	0.000	0.008	-0.121	0.058
silver_close_ret	0.000	0.017	-0.376	0.131
brent_close_ret	0.000	0.020	-0.280	0.191
dji_close_ret	0.000	0.009	-0.138	0.108
spx_close_ret	0.000	0.010	-0.128	0.091
rut_close_ret	0.000	0.012	-0.154	0.090
nasdaq_close_ret	0.000	0.012	-0.131	0.115
usdcny_close_ret	0.000	0.003	-0.022	0.021
usdeur_close_ret	0.000	0.004	-0.028	0.028
gold_volume_chg	0.001	1.648	-6.144	12.864
silver_volume_chg	0.002	1.668	-6.895	11.267
brent_volume_chg	0.000	0.470	-3.536	4.383
dji_volume_chg	0.001	0.194	-1.272	1.320
spx_volume_chg	0.000	0.144	-1.075	0.835
rut_volume_chg	0.000	0.144	-1.075	0.835
nasdaq_volume_chg	0.000	0.145	-2.453	2.376
vix_close_chg	0.000	0.067	-0.442	0.768
hash-rate_chg	0.002	0.122	-0.549	0.594
difficulty_chg	0.002	0.015	-0.243	0.163
median-confirmation-time_chg	0.000	0.299	-3.957	3.787
blockchain_log_mempool_count_chg	0.000	0.060	-0.254	0.248

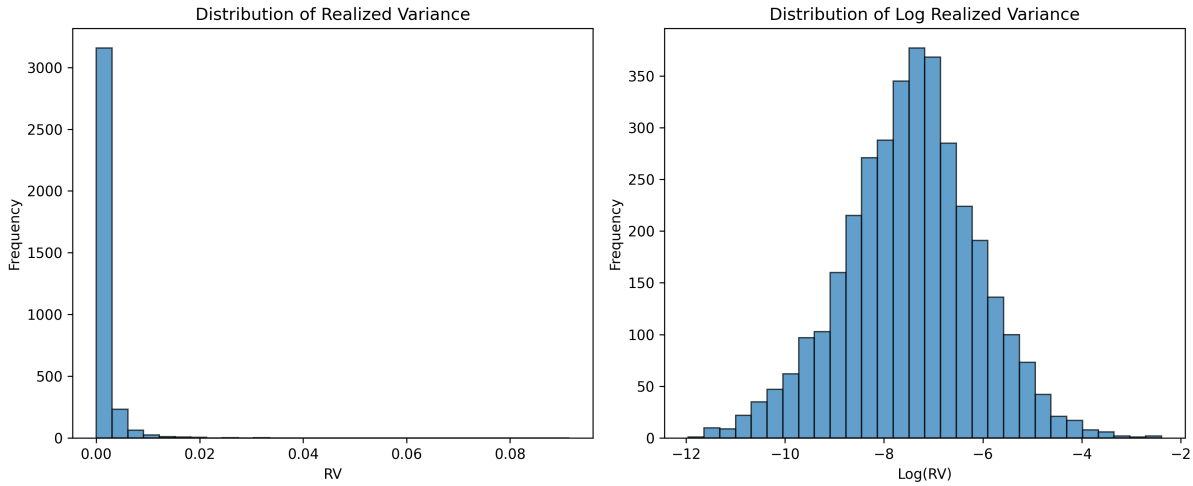


Figure 1: Distribution and autocorrelation diagnostics of realized variance. Left: Distribution of realized variance. Right: Distribution of log realized variance.

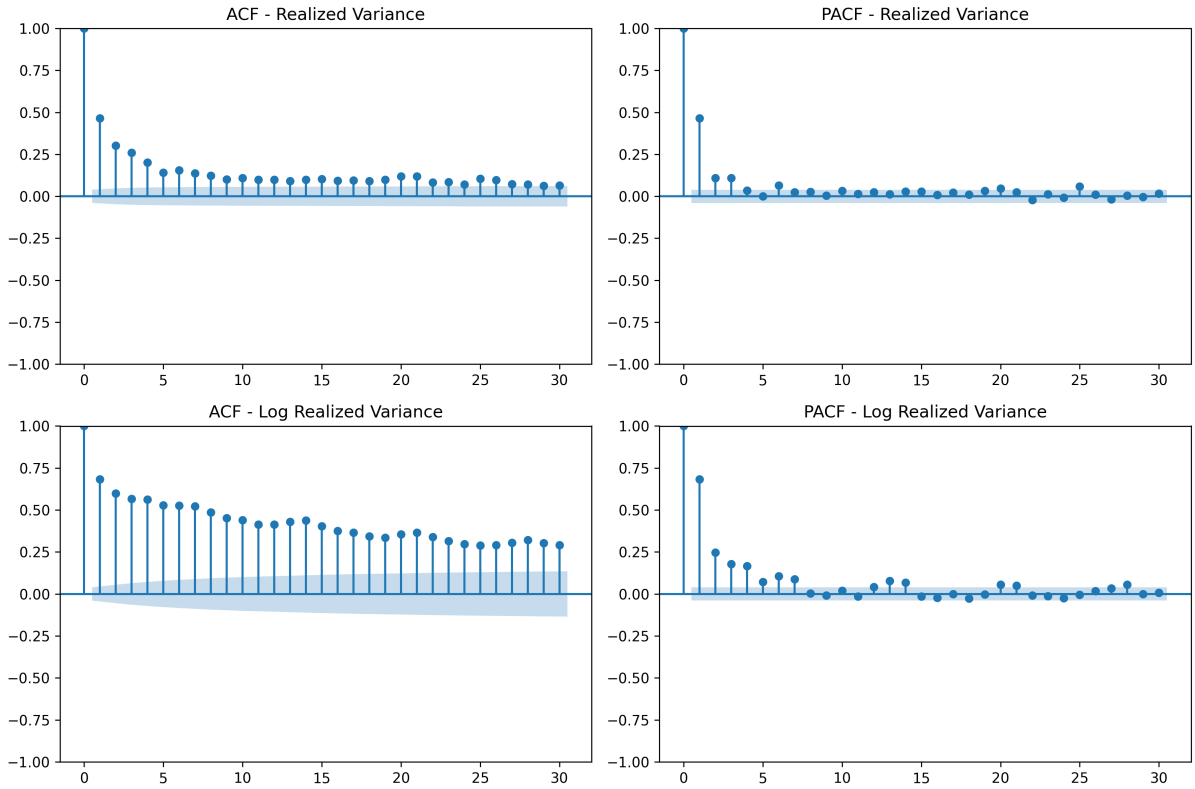


Figure 2: Autocorrelation diagnostics for realized variance and its logarithmic transformation. Top-left: ACF of realized variance. Top-right: PACF of realized variance. Bottom-left: ACF of log realized variance. Bottom-right: PACF of log realized variance.

The autocorrelation diagnostics in Figure 2 indicate strong persistence in realized variance and its logarithmic transformation. The ACF exhibits a slow decay, while the PACF shows significant spikes in the first few lags before tapering off. This supports the inclusion of short-term lags and a 30-day rolling mean to capture both short- and medium-term volatility dynamics.

Table 7: Class Distribution Under Different Regime Definitions

Regime	**Binary Count**	**Ternary Count**	**Ternary Proportion (%)**
High	1920	1151	32.97
Low	1569	1612	46.18
Distress	–	726	20.85
Total	3489	3489	100.00

Table 8: SVR Hyperparameter Grid

Parameter	**Values**
C	$\{0.1, 1.0, 10.0\}$
ϵ	$\{10^{-4}, 10^{-3}, 10^{-2}\}$

Notes: Bold values indicate selected hyperparameters based on validation performance (QLIKE).

Table 9: LSTM Hyperparameter Grid

Parameter	**Values**
lookback	$\{20, 30, 60\}$
units	$\{32, 64\}$
dropout	$\{0, 0.1, \mathbf{0.2}\}$
batch_size	$\{32, \mathbf{64}\}$

Notes: Bold values indicate selected hyperparameters based on validation performance (QLIKE).

Table 10: Random Forest Hyperparameter Grid

Parameter	**Values**
$n_{\text{estimators}}$	$\{\mathbf{500}, 1000\}$
max_depth	$\{\text{None}, 5, \mathbf{10}\}$
min_samples_leaf	$\{1, 5, \mathbf{10}\}$

Notes: Bold values indicate selected hyperparameters based on validation performance (QLIKE).

Weighted LSTM Objective

To account for class imbalance and the temporal structure of regime switching, we adopt a weighted training objective following Agakishiev et al. (2025). Each observation is assigned a weight w_t defined as

$$w_t = (1-\lambda)\frac{T}{\sum_{s=1}^{T} \mathbf{1}\{g_t = g_s\}} + \lambda \frac{\sum_{s=1}^{T} \mathbf{1}\{g_s \neq g_{s-1}\}}{\sum_{s=1}^{T} \mathbf{1}\{g_t = g_s\}\mathbf{1}\{g_s \neq g_{s-1}\}} \tag{4}$$

where T denotes the number of observations, g_t is the regime label at time t, and $\mathbf{1}\{\cdot\}$ is the indicator function. The parameter $\lambda \in [0, 1]$ controls the relative emphasis placed on regime-switching observations.

The LSTM is trained using a weighted categorical cross-entropy loss given by

$$\mathcal{L} = -\frac{1}{T}\sum_{t=1}^{T} w_t \sum_{k=1}^{K} y_{t,k}\log(\hat{y}_{t,k}), \tag{5}$$

where K denotes the number of regimes, $y_{t,k}$ is the true indicator for regime k at time t, and $\hat{y}_{t,k}$ is the predicted probability from the LSTM softmax layer.

This weighting scheme increases the contribution of observations associated with regime transitions, thereby improving the model's ability to capture regime dynamics.

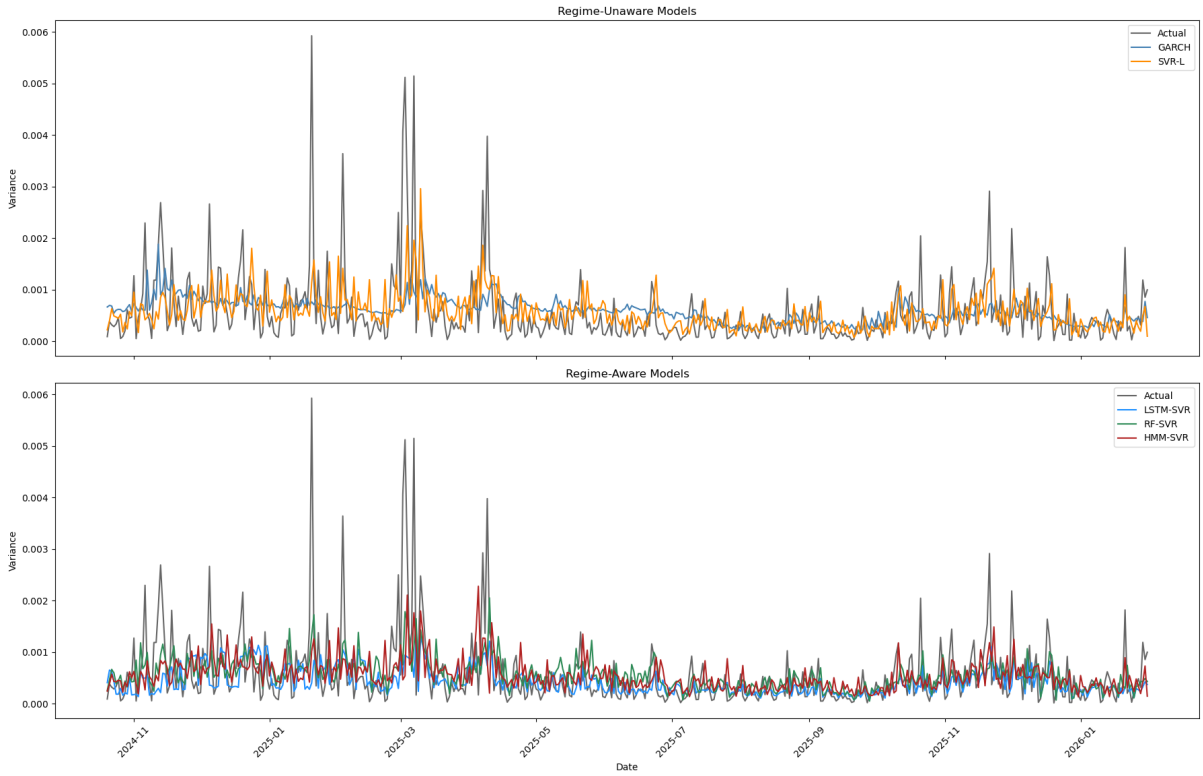
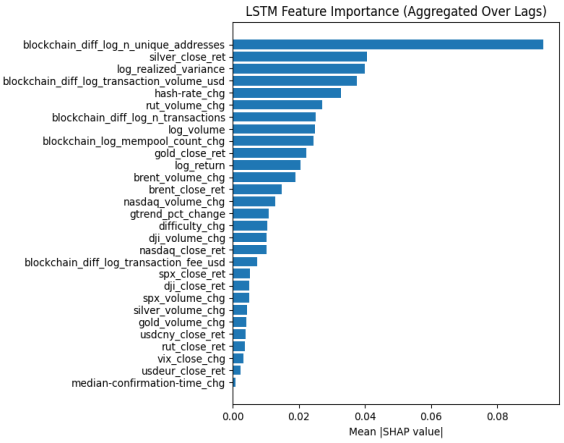


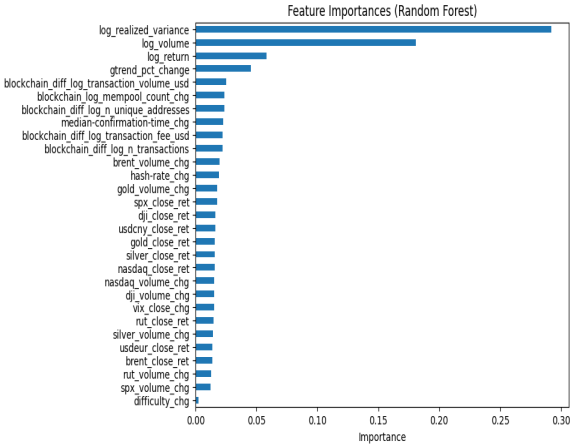
Figure 3: Comparison of actual and predicted realized variance over the test period. Top panel: regime-unaware models (GARCH, SVR-L). Bottom panel: regime-aware models (LSTM-SVR, RF-SVR, HMM-SVR). All models tend to underestimate volatility spikes, with regime-aware models tracking closer to actual variance during transitional periods.

Table 11: Comparison of classification performance across supervised models

Model	Class	Precision	Recall	F1-score	Support
LSTM	Low	0.56	0.64	0.60	214
	High	0.66	0.58	0.62	255
	Accuracy		0.61		469
	Macro Avg	0.61	0.61	0.61	469
	Weighted Avg	0.61	0.61	0.61	469
Random Forest	Low	0.53	0.73	0.62	214
	High	0.67	0.46	0.55	255
	Accuracy		0.59		469
	Macro Avg	0.60	0.60	0.58	469
	Weighted Avg	0.61	0.59	0.58	469

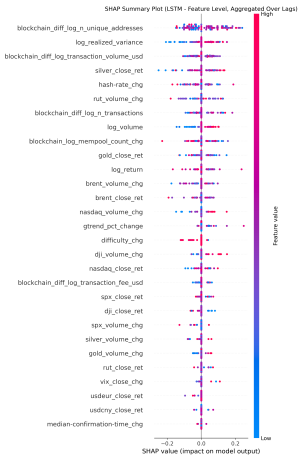


(a) LSTM Feature Importance

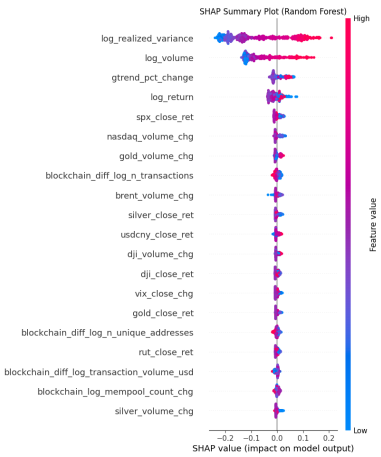


(b) Random Forest Feature Importance

Figure 4: Comparison of feature importance across models



(a) LSTM SHAP values



(b) Random Forest SHAP values

Figure 5: Comparison of SHAP-based feature importance across models

Table 12: Out-of-sample forecasting performance across regime-aware models

Model	RMSE	MSE	MAE	QLIKE
LSTM–SVR	0.000666	4.4395×10^{-7}	0.000368	0.525050
RF–SVR	0.000609	3.7129×10^{-7}	0.000374	0.385672
HMM–SVR	0.000646	4.1784×10^{-7}	0.000388	0.451964
Neutral–SVR	0.000626	3.9196×10^{-7}	0.000367	0.396909
Oracle–SVR	0.000595	3.5416×10^{-7}	0.000287	0.259427

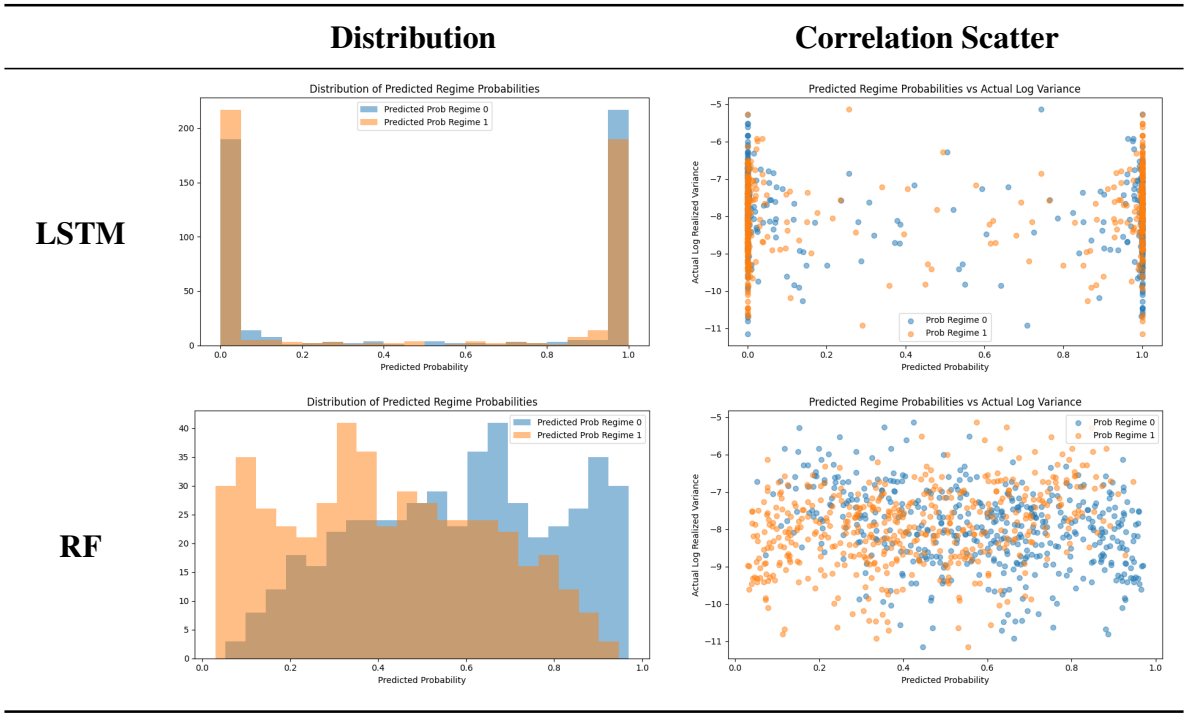
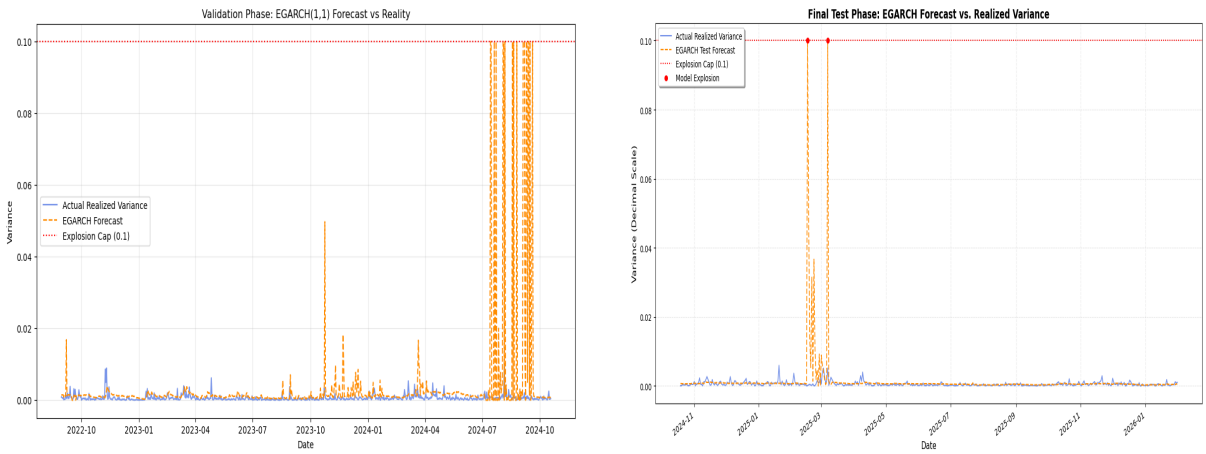


Figure 6: Comparison of predicted regime probability distributions and their relationship with log realized variance for the LSTM and Random Forest models.



(a) EGARCH Predictions

(b) EGARCH Instability

Figure 7: EGARCH Instability Forecasts

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